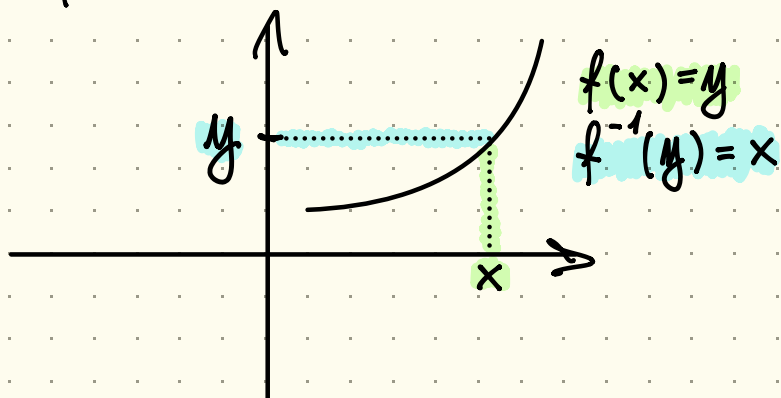


RIPASSO

$$f: A \rightarrow \mathbb{R} \quad \text{INIETTIVA}$$



es) $f(x) = x + 1 \quad y = x + 1$

$$f^{-1}(x) = y - 1 = x \rightsquigarrow f^{-1}(x) = x - 1 = y$$

• SE LA FUNZIONE DI PARTENZA È:

$$y = x + 1$$



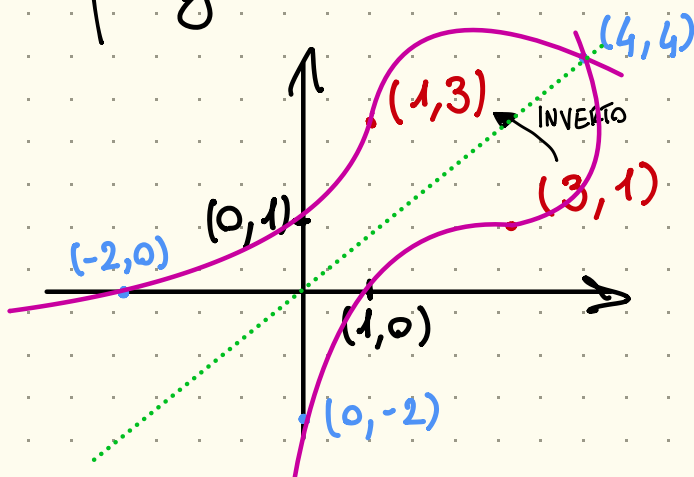
$x \leftrightarrow y$ SCAMBIO x E y
(INPUT, OUTPUT)

$$x = y + 1$$



$$f \quad y = x - 1$$

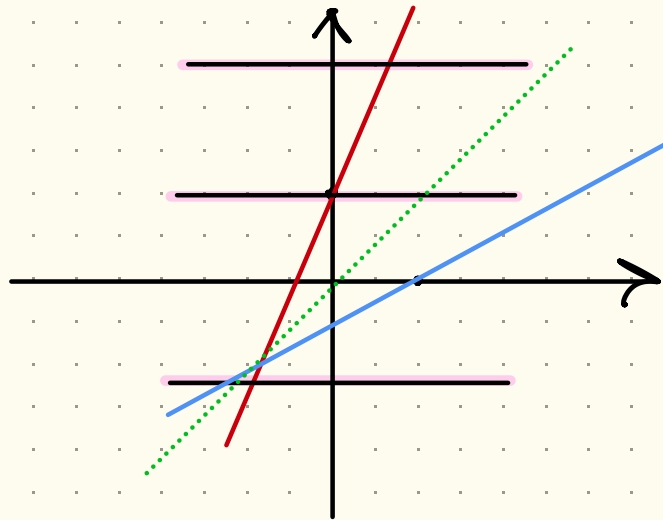
SE HO UNA f CON
INPUT 4 E OUTPUT 4
INVERTENDO AVRÒ
SEMPRE INPUT 4 E
OUTPUT 4



b) $f(x) = 2x + 2$

1. CONTROLLARE SE È INIETTIVA:

LA TAGLIO ORIZZONTALMENTE CON RETTE
E VERIFICO CHE NON CI SIA MAI PIÙ
DI UN PUNTO DI INTERSEZIONE



$$y = 2x + 2$$

$$\Downarrow \quad x \leftrightarrow y$$

$$x = 2y + 2$$

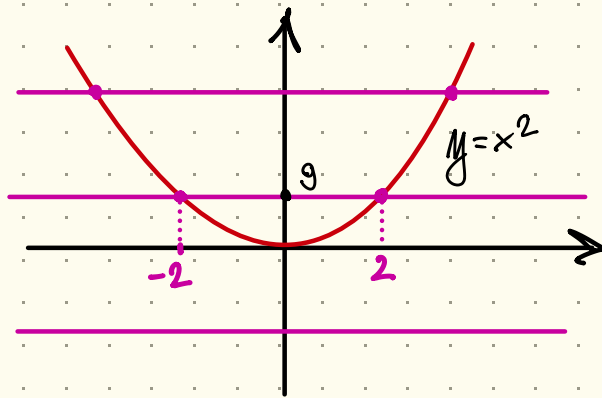
$$\Downarrow$$

$$x - 2 = 2y$$

$$\Downarrow$$

$$y = \frac{x-2}{2} \quad f^{-1}$$

es) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

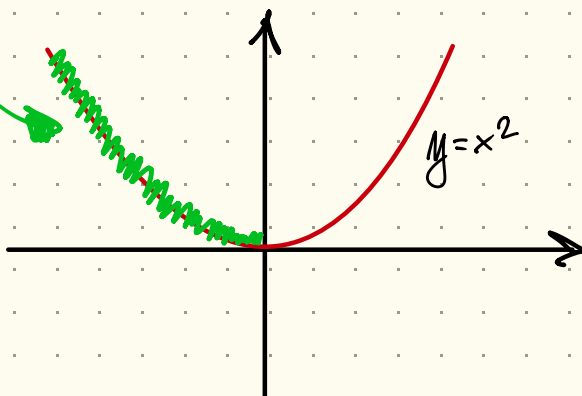


NON INIETTIVA
(PER OGNI INPUT HO PIÙ OUTPUT)

$$\begin{aligned} y &= x^2 \\ \Downarrow \\ x &= y^2 \\ \Downarrow \text{?} \\ y &= \begin{cases} \sqrt{x} \\ -\sqrt{x} \end{cases} \end{aligned}$$

• RESTRINGO IL DOMINIO: $[0, +\infty)$

TOLGO
LA PARTE
SINISTRA
DELLA FUN.



ORA È INIETTIVA!! → POSSO FARE L'INVERSA

L'INVERSA DI $f|_{[0, +\infty)}$ È

• $\sqrt{\cdot}: [0, +\infty) \rightarrow \mathbb{R}$

• $\sqrt{9} = 3$

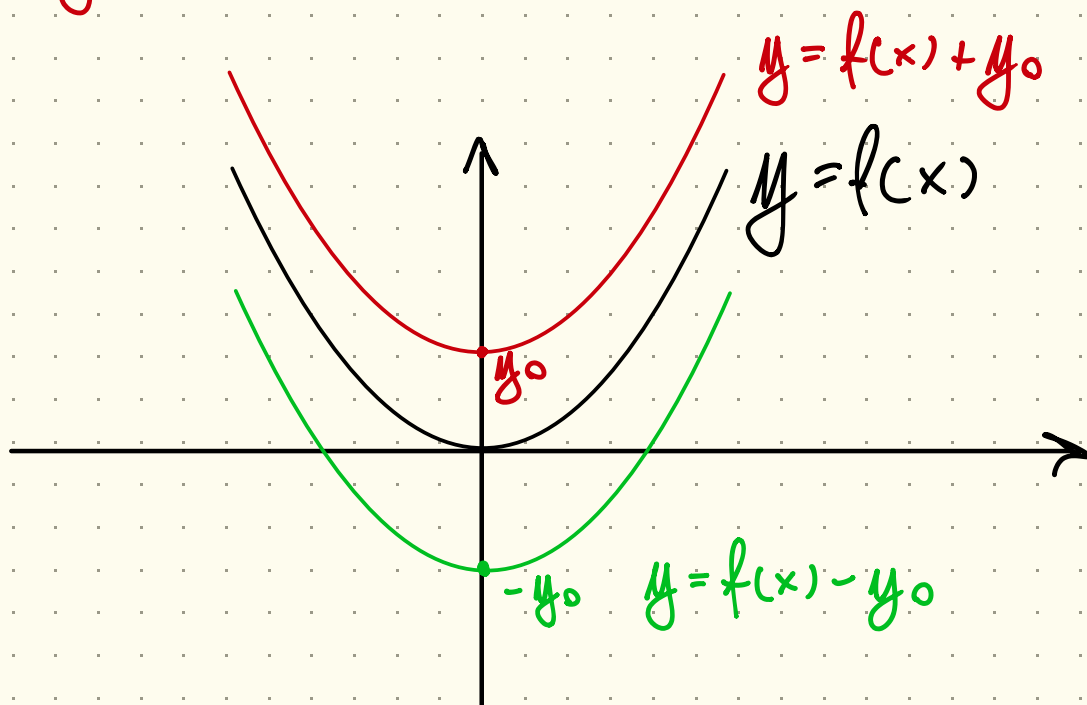
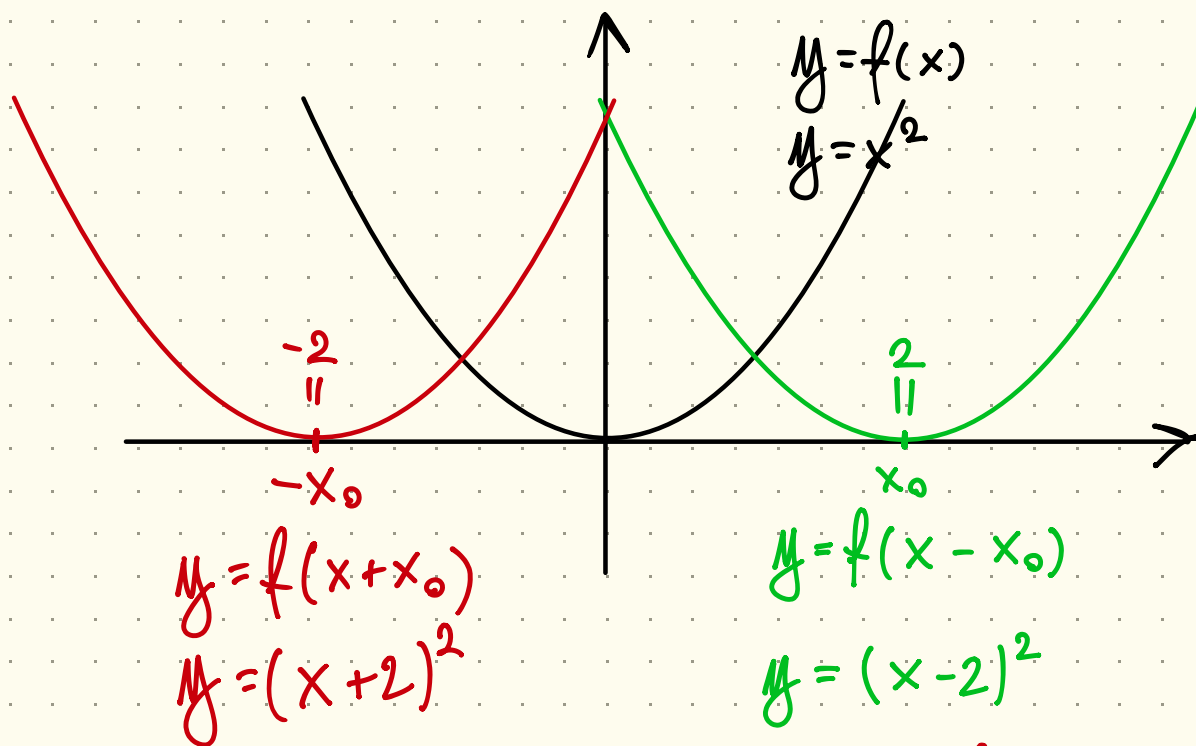
$$\sqrt{x^2} \neq x \quad f^{-1} \circ f \neq \text{identità}$$

$$\sqrt{(-3)^2} = 3$$

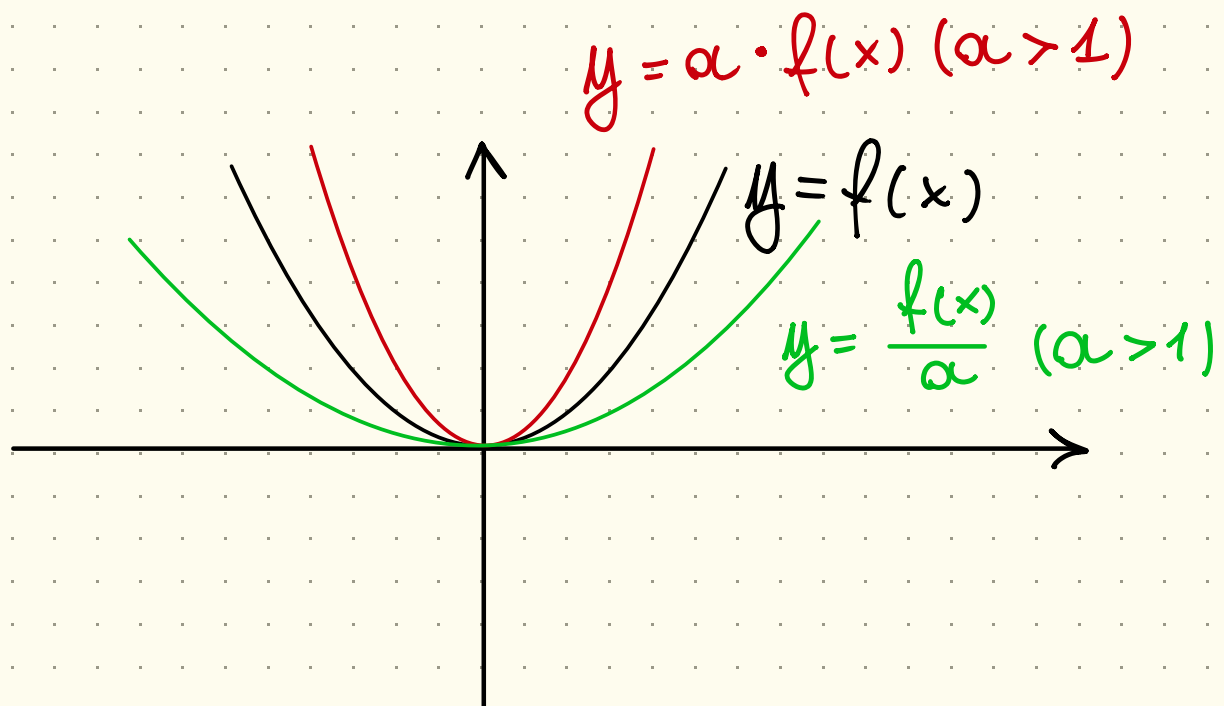
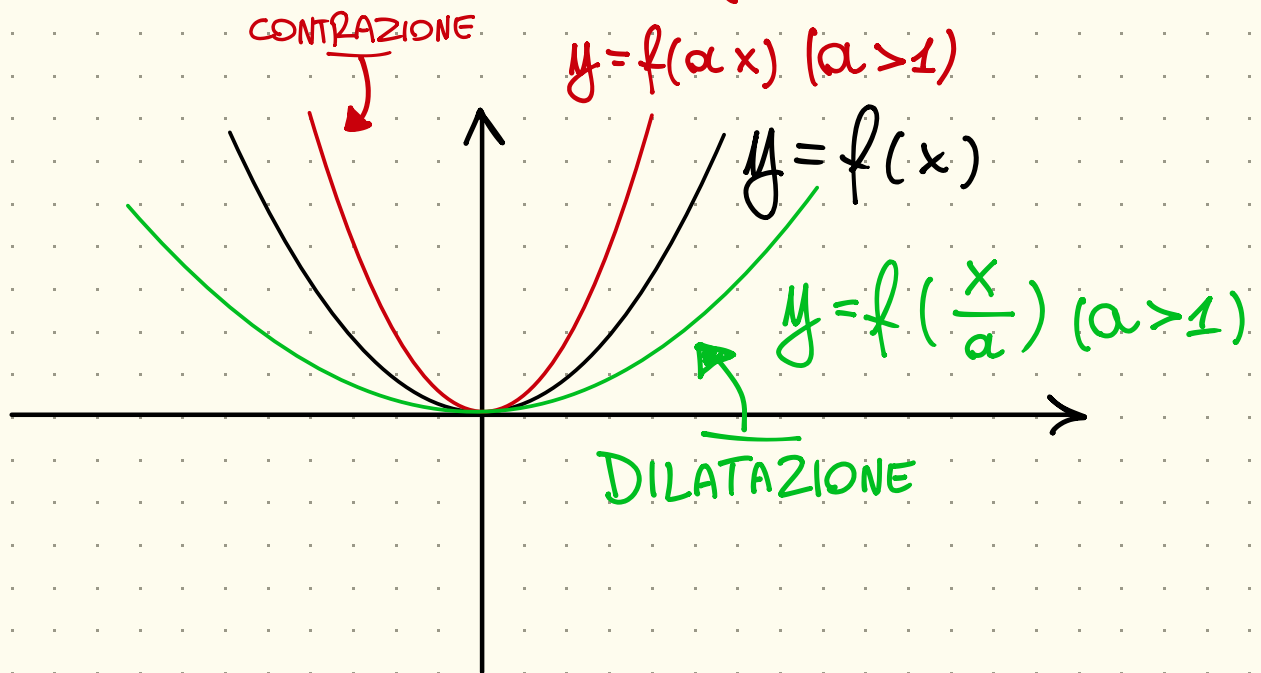
$$(\sqrt{x})^2 = x \quad f \circ f^{-1} = \text{id}$$

TRASFORMAZIONI DI GRAFICI

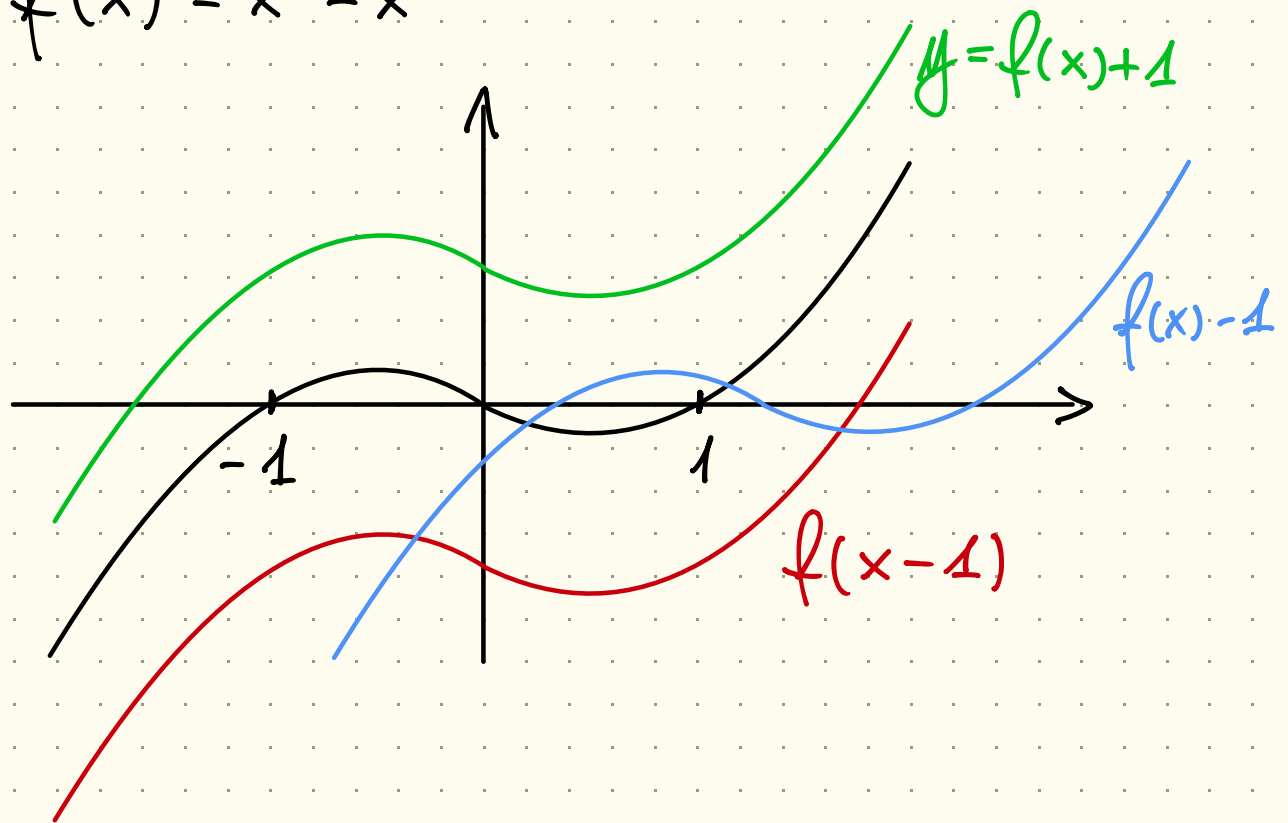
• TRASLAZIONI (+/-)



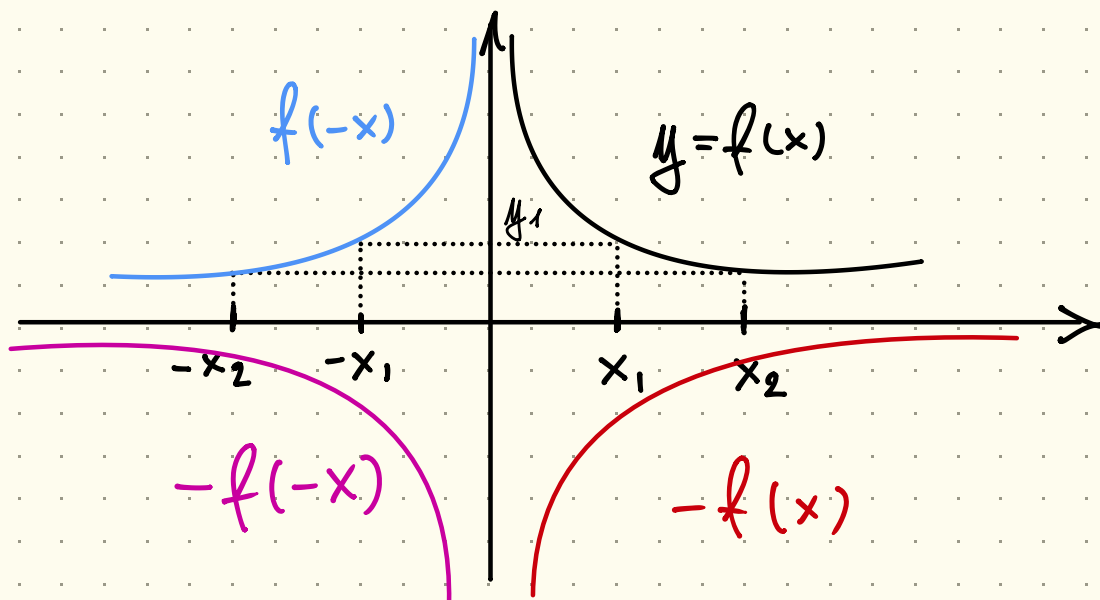
• PRODOTTO E QUOZIENTE ($\times / \div$)



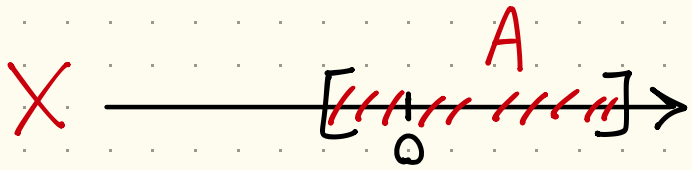
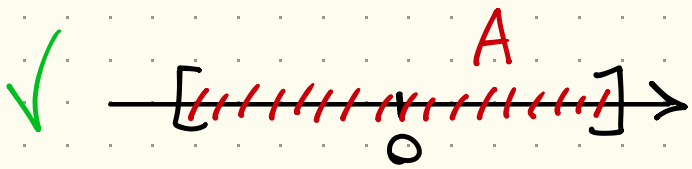
es) $f(x) = x^3 - x$



• SIMMETRIE



DEF. $f: A \rightarrow \mathbb{R}$ con A t.c.
 $x \in A \Leftrightarrow -x \in A$
 (A simmetrico)



f SI DICE PARI SE:

$$\forall x \in A \quad f(x) = f(-x)$$

